

OMAR WASEEM

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building multi-agent LLM workflows, RAG-grounded retrieval systems, and production-style data pipelines across AWS and Azure. Designed a LangGraph/LangChain vendor-risk system with human-approval gating at Cognizant and built a serverless summarization pipeline on Bedrock with sub-2s response times. Computer science graduate pursuing a master's in AI & Machine Learning, focused on building AI systems that are grounded, tested, and production-ready.

EDUCATION

Rutgers University - New Brunswick | May 2025

Bachelor of Science, Computer Science

Western Governors University | Expected May 2027

Master of Science, AI & Machine Learning

TECHNICAL SKILLS

- **Core Competencies:** System Design, Distributed Systems, Microservices, REST APIs, Object-Oriented Programming, Algorithms & Data Structures, Unit Testing
- **Languages:** Python, Go, C#, Java, HTML, JavaScript, CSS, C/C++, TypeScript
- **AI / ML:** LangChain, LangGraph, RAG, TensorFlow, Keras, Bedrock, Sagemaker, Neural Networks
- **Cloud & Data:** AWS (Lambda, API Gateway, DynamoDB, RDS, CloudWatch), Azure, Docker, Apache Kafka, Apache Hadoop, PySpark, MongoDB, PostgreSQL, SQL, SQLite, ETL, GitHub Actions, AWS EMR, Hive, Spark SQL
- **Frameworks:** React Native, Streamlit, Flask, Chart.js, AWS SAM, JWT Authentication, CI/CD Pipelines

EXPERIENCE

Cognizant | AI Engineer Intern | May 2026 - Aug 2026

- Designed a multi-agent LangGraph/LangChain workflow with RAG-grounded policy retrieval and a human-approval gate for vendor risk review, resulting in a Streamlit demo passing all 5 required test scenarios, including missing-document handling, a revision loop, and human escalation.
- Collaborated on a pooled LSTM model with per-stock embeddings and engineered features (RSI, MACD, volatility) from live market data across ~100 large-cap stocks, resulting in a capacity-ablation study spanning 3K-75K parameters that confirmed the model's accuracy ceiling was driven by limited signal in the data, not model size.
- Built and validated a config-driven Python ETL pipeline for flight, hotel, and weather data, delivering a PostgreSQL-backed reporting system covering 98 cities with 87% test coverage across 33 tests.

Revature | Data Engineering Intern | Feb 2026 - May 2026

- Built a Python capstone pipeline ingesting, validating, and loading CSV data into a SQLite database, producing a fully queryable analytics layer for SQL-based reporting.
- Implemented batch and distributed processing with PySpark (RDDs, DataFrames, Spark SQL), building scalable ingestion and transformation workflows validated on AWS EMR clusters.
- Built ingestion, transformation, and storage workflows across the Hadoop ecosystem (HDFS, Hive, Spark) for reliable daily reporting at scale.
- Completed a 9-week intensive data engineering curriculum (SQL, Git, Python, Hadoop, Hive, PySpark) with 58 coding labs and an end-to-end capstone.

Xtremeanalytix | Cloud Engineering Intern | Jun 2024 - Aug 2024

- Designed and deployed five data pipelines across SQL Server, MongoDB, AWS RDS, and Azure SQL in cloud and hybrid environments, accelerating data integration for downstream applications.
- Cut data ingestion latency 40% across 3 cloud-hosted applications by building a custom MongoDB source connector in Java that streams data in real time into Apache Kafka.
- Reduced pipeline errors 30% by optimizing data synchronization across 4+ cloud-native and on-prem systems.

TECHNICAL PROJECTS

Multi-Agent Vendor Onboarding & Risk Review | Jul 2026 - Aug 2026 | <https://github.com/OWaseem/Multi-Agent-Vendor-Onboarding-Risk-Review>

- Built a model-agnostic LLM client with a deterministic template fallback, so the full workflow runs and passes every test scenario with zero API keys configured.
- Implemented PDF document validation against per-document-type field schemas, automatically rejecting malformed or incomplete vendor documents such as expired certificates and malformed tax IDs.

Serverless AI Text Summarization Pipeline | Feb 2026 - Apr 2026 | <https://github.com/OWaseem/Serverless-AI-Text-Summarization-Pipeline>

- Architected a production-style serverless AI pipeline using AWS SAM, API Gateway, and AWS Lambda, with results persisted in DynamoDB and all endpoints secured via API key authentication, achieving an average retrieval time of 0.72s.
- Integrated Claude Haiku 4.5 via Amazon Bedrock to automate text summarization and sentiment tagging, reducing long-form text by up to 61% with automated positive/negative/neutral classification applied to every processed document.
- Developed a fully event-driven, auto-scaling system with CloudWatch logging, achieving zero server maintenance and an average end-to-end response time of 1.89s.

Pooled LSTM Multi-Stock Return Forecaster | Jun 2026 - Jul 2026 | https://github.com/OWaseem/AI_project_1

- Benchmarked the model against three baselines: zero-return, persistence, and gradient-boosted trees, reporting an honest result of no meaningful directional edge rather than an inflated one.
- Built a Flask web app serving live next-day predictions through an auto-refreshing UI that runs without retraining the model.

Travel Summary ETL Pipeline | May 2026 - Jun 2026 | https://github.com/OWaseem/Revature_ETL_Data_Pipeline

- Integrated the Open-Meteo API to pull live weather per arrival city, joining it with flight and hotel records into a unified reporting view across 10 global regions.
- Built a Flask and Chart.js dashboard with a searchable rejects table and pipeline-health charts, giving a self-service view into pipeline stats without direct database access.

PEAKMODE – AI Health Supplement Recommender App | Mar 2025 - May 2025 | <https://github.com/OWaseem/PEAKMODE>

- Designed and built an AI recommendation engine using Bayesian reasoning and informed search, producing personalized supplement suggestions with confidence scores at a 250ms response time.
- Built the React Native mobile frontend with JWT authentication, SQLite storage, and dark mode, ensuring a polished cross-platform experience on iOS and Android.
- Owned the project end-to-end architecture through deployment across the AI backend, mobile frontend, auth, and data layer.

CERTIFICATIONS & AWARDS

- AWS Certified AI Practitioner - AWS: Issued 2026
- Machine Learning Specialization - DeepLearning.AI: Issued 2026
- Fundamentals of AI Agents Using RAG and LangChain - IBM: Issued 2026
- Eagle Scout - Boy Scouts of America: Rank earned by fewer than 6% of Scouts worldwide: Issued 2021